

5(a)(i)	A = glomerulus ; B = Bowman's capsule ;	2
5(a)(ii)	any four from: 1 ultrafiltration ; 2 high(er) pressure due to afferent arteriole wider than efferent arteriole ; 3 capillary endothelium has, fenestrations / gaps / holes / pores ; 4 small molecules / (named) ions / glucose / amino acids / urea / water, and EITHER leave, blood / plasma / capillary / glomerulus OR enter, capsule / filtrate / tubule ; 5 basement membrane forms, (main) filtration / selective, barrier ; 6 (R)MM / M_r, limit of, 68 000 to 70 000 (Da) / 68–70 kDa ; 7 podocytes, form slit pores / support basement membrane ;	4
5(b)(i)	C adenylyl cyclase ; D cyclic AMP ;	2
5(b)(ii)	any four from: 1 in both as Na⁺ concentration increases ADH concentration increases (after a, certain / stated, point) ; 2 in, type C / SIAD, ADH rise occurs at a lower Na⁺ concentration ; 3 comparative figures ; 4 (ADH →) more aquaporins / greater permeability to water, in collecting duct / DCT ; 5 (C / SIAD / ADH →) more water reabsorbed, at collecting duct / at DCT / into blood ; 6 (C / SIAD / ADH →) small(er) volume of / (more) concentrated, urine ; 7 type C / SIAD, causes, high blood water potential / low blood Na⁺ concentration ;	4

Question	Answer	Marks
----------	--------	-------